alsoft

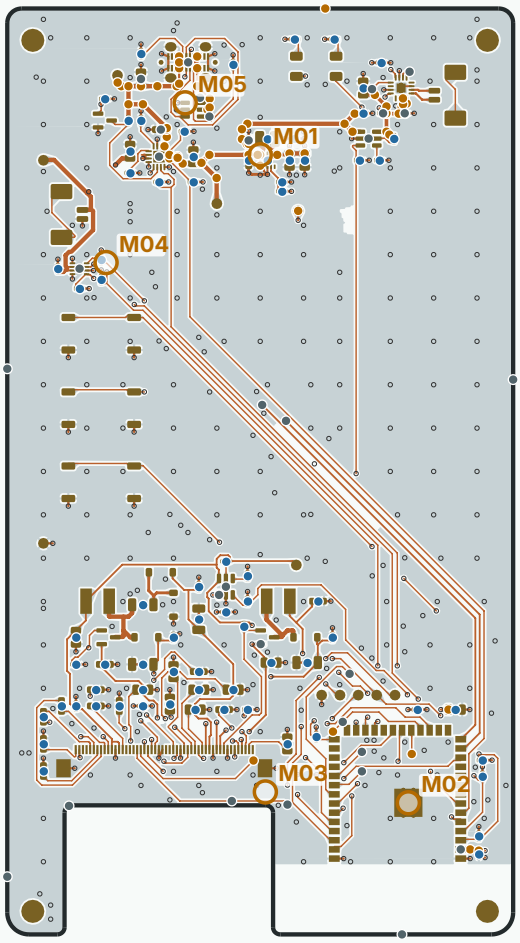
CONFIDENTIAL

Paper Calendar - ESP32-S3 e-paper board (PhotoPainter drop-in)

1. Cover

Customer	alsoft (own board)
Board	Paper Calendar - ESP32-S3 e-paper board (PhotoPainter drop-in)
Revision	0.1
Date	2026-11-01
Reviewer	alsoft

0	26	1	80
High	Medium	Low	Info



Top side (F.Cu). Labelled markers are the priority findings; dots are further findings. The bottom side and every close-up follow in the board map sections.

Priority findings

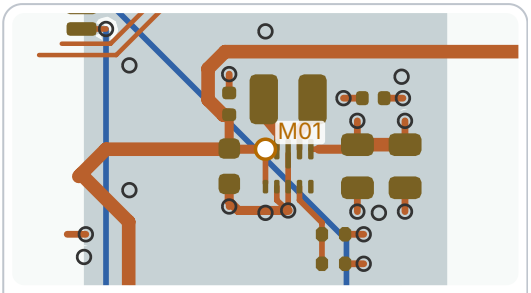
The 5 highest-risk findings of 107. Every card, close-up and record follows in the numbered sections.

M01 decoupling_order

medium · U3.10, C7, C6

U3 pin 10 (VSYS): the small 100 nF capacitor C7 is 2.2 mm away while the larger 10 µF C6 is closer at 1.6 mm, so the small one is not nearest the pin. Fix: swap them.

Full finding M01



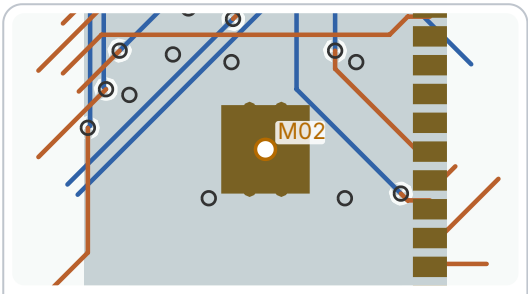
M01 close-up

M02 exposed_pad_vias

medium · U5.41

U5: the 3.9 x 3.9 mm exposed pad has 10 vias where 11 are recommended for heat and ground. Fix: add a via grid inside the pad.

Full finding M02



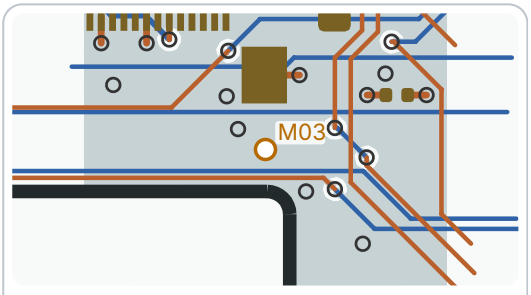
M02 close-up

M03 edge_clearance

medium · J2

J2: the part sits 1.85 mm from the board edge but assembly needs 2.5 mm. Fix: move it inward, or order with edge rails at extra cost.

Full finding M03



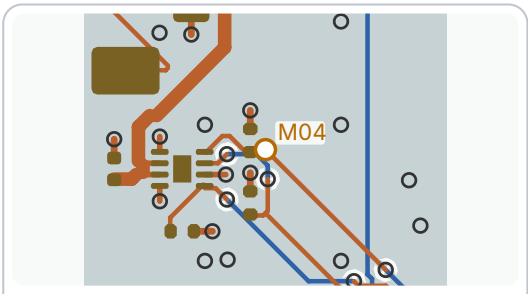
M03 close-up

M04 acute_angles

medium · I2C_SDA

I2C_SDA: a track corner on F.Cu is sharper than recommended (tracks meet at 45 degrees), which can cause etching defects. Fix: re-route with gentler turns.

Full finding M04



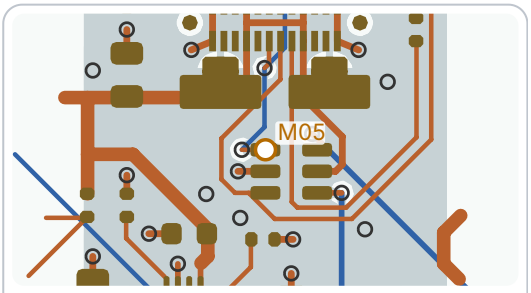
M04 close-up

M05 esd_order

medium · J1.A7, J1.B7, U1.1, USB_C_DN

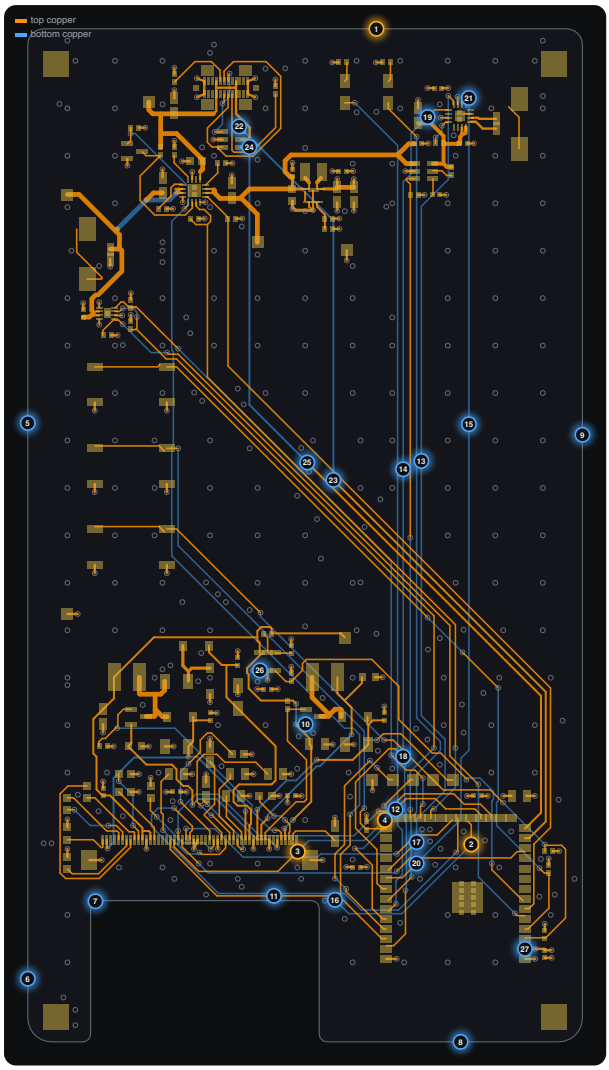
USB_C_DN: the protection part U1 sits 2 via(s) away from connector pin J1 pin A7, so surges can reach the chip first. Fix: place it right behind the connector pad on the same layer. The same issue appears at 1 more place.

Full finding M05



M05 close-up

EMC overview



Whole-board EMC map. Marker numbers follow the EMC check's own order; report IDs are listed at the right.

Top EMC findings

M23 edge_stitching

A 42 mm stretch of the board edge has no ground via (limit 20 mm), so the plane pair can radiate there. Fix: add a ground via fence along that edge.

Full finding M23

M24 layer_change

EPD_MOSI: the signal changes layers (F.Cu to B.Cu) 6.2 mm from its nearest return-current path, which can radiate. Fix: add a stitching capacitor between the planes next to the via.

Full finding M24

M25 layer_change

EPD_SCK: the signal changes layers (F.Cu to B.Cu) 7.1 mm from its nearest return-current path, which can radiate. Fix: add a stitching capacitor between the planes next to the via.

Full finding M25

M26 layer_change

EPD_SCK: the signal changes layers (F.Cu to B.Cu) 6.9 mm from its nearest return-current path, which can radiate. Fix: add a stitching capacitor between the planes next to the via.

Full finding M26

I12 edge_stitching

A 107 mm stretch of the board edge has no ground via (limit 20 mm), so the plane pair can radiate there. Fix: add a ground via fence along that edge.

Full finding I12

Inputs

No reviewer input list supplied; import artifacts identify the work CAD.

Independent review of the supplied design and newly generated evidence. Existing customer test results are not used. An optional bench appendix is added only after the review and does not alter findings.

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2. Summary

0 High findings	**26** Medium findings	**1** Low findings
80 Info observations	**79** Coverage gaps	**9** Schematic pages read
8 Steps run	**8** Steps not run	

This independent review contains 107 ranked findings: 0 high, 26 medium, 1 low and 80 informational. 8 steps ran and 8 did not run; incomplete checks remain explicit below. 9 schematic pages were exported and included as review evidence; this count does not claim human sign-off. KiCad DRC confirmations are shown separately and do not increase the ranked finding counts. Resolve the findings and complete missing evidence before engineering release.

79 coverage groups are excluded from finding counts. What was not checked, why, and the inputs needed to close each gap appear in Tool limitations. Every original producer record and location is retained in Appendix A. This report requires reviewer adjudication and is not a release sign-off.

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3. Findings

The first 12 priority cards and close-ups are shown here. The complete finding index, remaining cards and locations are in Appendix A; counts include all findings.

Complete finding index

M01 — decoupling_order

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: U3.10, C7, C6

Why: affected location (100n) is 2.2 mm from affected location [VSYS], affected location (10u) 1.6 mm

Recommendation: swap the capacitors' places, or move the small one next to the pin

Source (step / check): board / boardkit review / decoupling_order: of the capacitors on an IC supply pin, the smallest value sits closest to the pin (it supplies the fastest current steps; bulk capacitors can be further away)

All 1 producer records and locationsBoard-map marker M01 · Close-up M01

M02 — exposed_pad_vias

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: U5.41

Why: U5 exposed pad 3.9 x 3.9 mm (15.21 mm^2) has 10 vias, 11 recommended

Recommendation: a via grid in the pad (fanout_planes exposed_in_pad=True; JLC_VIA_SMALL fits a 2x2 grid in small pads)

Source (step / check): board / boardkit review / exposed_pad_vias: an exposed (heatsink) pad has at least one via per ReviewOptions.ep_area_per_via mm^2 to the plane (electrical and thermal path)

All 1 producer records and locationsBoard-map marker M02 · Close-up M02

M03 — edge_clearance

Severity: medium

Medium: assembly edge handling requires a process change or rails; this is not proof the board cannot be fabricated.

Where: J2

Why: Assembled parts are inside the required assembly edge margin; per-location distances and limits are in the appendix.

Recommendation: move the part in, or order with edge rails (>= 5 mm; JLC adds them by default and charges for the panel)

Source (step / check): board / boardkit review / edge_clearance: assembled parts (body and pads) keep ReviewOptions.edge_clearance (JLC: 2.5 mm) from the board edge, or the order needs edge rails

All 1 producer records and locationsBoard-map marker M03 · Close-up M03

M04 — acute_angles

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: I2C_SDA

Why: affected location on F.Cu: tracks meet at 45 degrees

Recommendation: re-route the corner with 45 degree turns, or open it up so the tracks meet at 90 degrees or more

Source (step / check): board / boardkit review / acute_angles: no track corner or junction encloses less than 90 degrees (acid traps)

All 1 producer records and locationsBoard-map marker M04 · Close-up M04

M05 — esd_order

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: J1.A7, J1.B7, U1.1, USB_C_DN

Why: affected location: 2 via(s) between affected location and the protection part affected location

Recommendation: put the protection part on the signal path right behind the connector pad, connected on one layer, and route on from it to the IC

Source (step / check): board / boardkit review / esd_order: on a net joining a connector, a protection part and an IC, the connector's copper reaches the protection part first (no branch of more than 0.5 mm) and without a via

All 2 producer records and locationsBoard-map marker M05 · Close-up M05

M06 — current_override

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: 3V3

Why: class PWR_050 wins: 0.500 mm track / 0.200 mm clearance; calculation 0.502 / 0.100 mm

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_override: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M07 — current_override

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: AMP_VDD

Why: class PWR_050 wins: 0.500 mm track / 0.200 mm clearance; calculation 0.603 / 0.100 mm

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_override: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M08 — current_override

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: VBUS

Why: class PWR_060 wins: 0.600 mm track / 0.200 mm clearance; calculation 1.435 / 0.100 mm

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_override: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M09 — current_pour

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: VBUS

Why: 1.500000 A needs 1.435 mm, beyond standard tracks; require a pour/plane and verify its bottlenecks

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_pour: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M10 — current_override

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: VSYS

Why: class PWR_060 wins: 0.600 mm track / 0.200 mm clearance; calculation 1.689 / 0.100 mm

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_override: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M11 — current_pour

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: VSYS

Why: 1.687739 A needs 1.689 mm, beyond standard tracks; require a pour/plane and verify its bottlenecks

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_pour: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 1 producer records and locations

No board-map marker: reference-only or board geometry unavailable.

M12 — current_width

Severity: medium

Medium: producer evidence indicates a likely defect or assembly yield/EMC risk under its stated assumptions.

Where: 3V3

Why: track (see appendix) on F.Cu: 0.300 mm for 0.700000 A; conservative chart requires 0.502 mm at 10.0 C rise (0.0350 mm copper); not a qualified short pad neck

Recommendation: Correct the copper or intent; document any exception with a current-stage waiver and reason

Source (step / check): current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper; current / boardkit current / current_width: Intent current, stackup copper and standard widths; IPC-2152 conservative chart, historical internal-curve approximation with 5% area allowance, both inner and outer copper

All 7 producer records and locationsBoard-map marker M12 · Close-up M12

KiCad cross-check confirmations

No matching KiCad confirmations were produced; unmatched DRC candidates remain in the DRC artifacts for engineering review.

Next steps

M01: swap the capacitors' places, or move the small one next to the pin

M02: a via grid in the pad (fanout_planes exposed_in_pad=True; JLC_VIA_SMALL fits a 2x2 grid in small pads)

M03: move the part in, or order with edge rails (>= 5 mm; JLC adds them by default and charges for the panel)

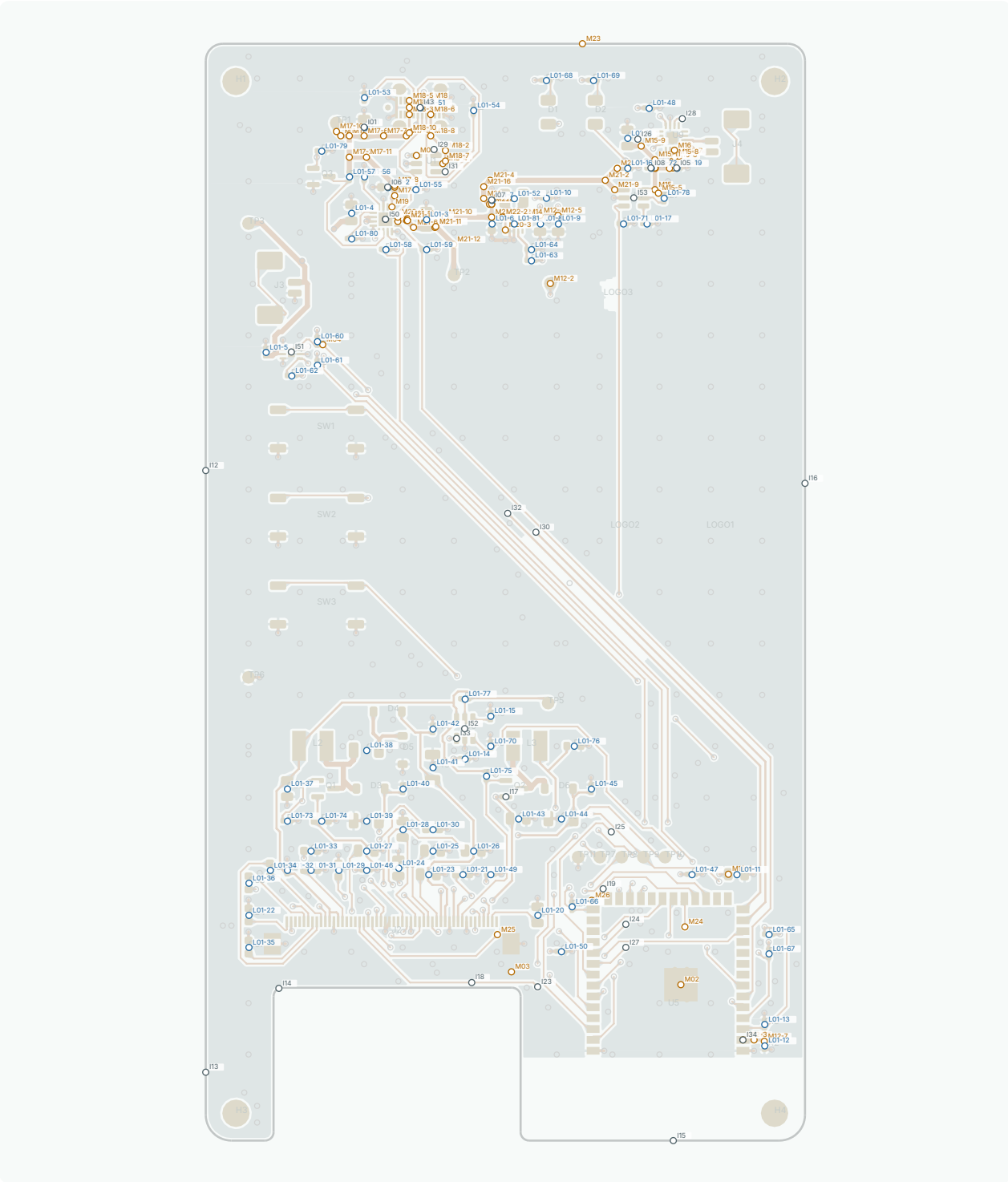
M04: re-route the corner with 45 degree turns, or open it up so the tracks meet at 90 degrees or more

M05: put the protection part on the signal path right behind the connector pad, connected on one layer, and route on from it to the IC

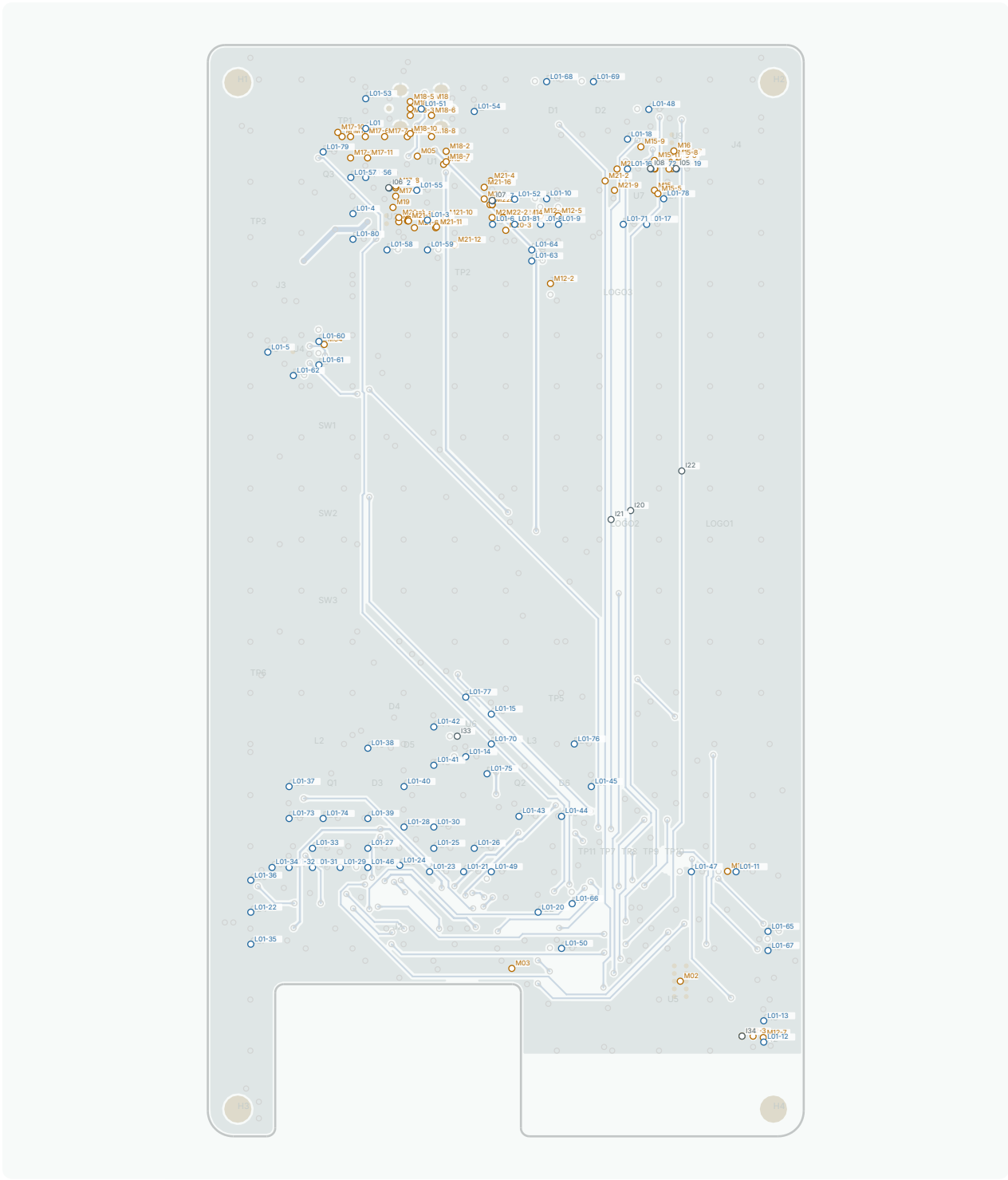
Address the priority findings above, then the remaining cards. Complete the coverage gaps and review incomplete steps before release; full reasons and records are in Appendix A.

4. Board map

Coordinates are input board-file millimetres, y down. Both views retain that frame. F/B findings appear on their side; unspecified and internal layers appear on both views with their layer noted in the finding.

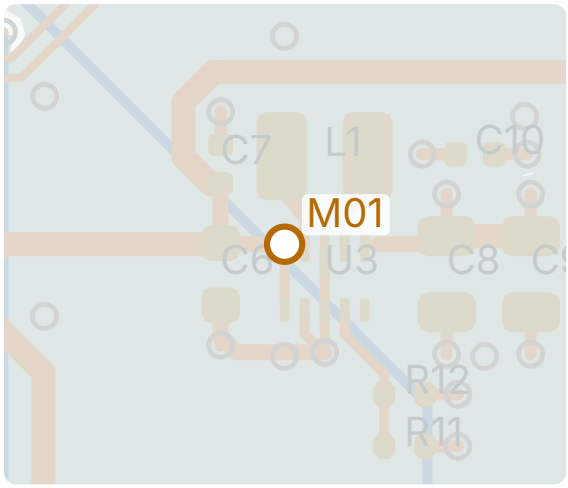


top — F.Cu

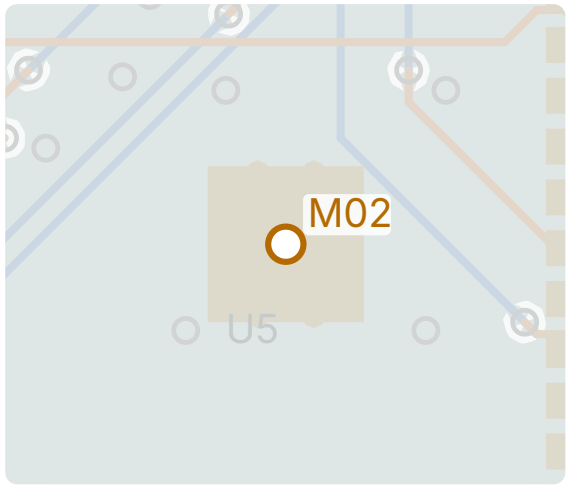


bottom — B.Cu

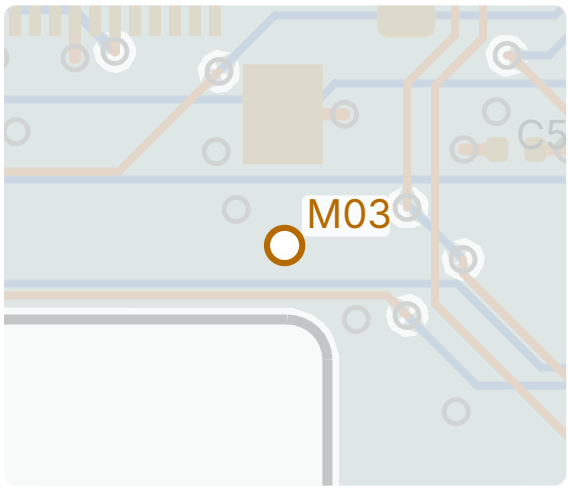
5. Close-ups



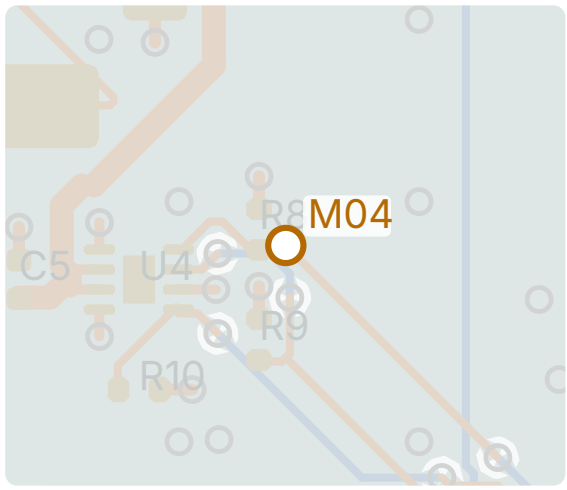
M01: decoupling_order —



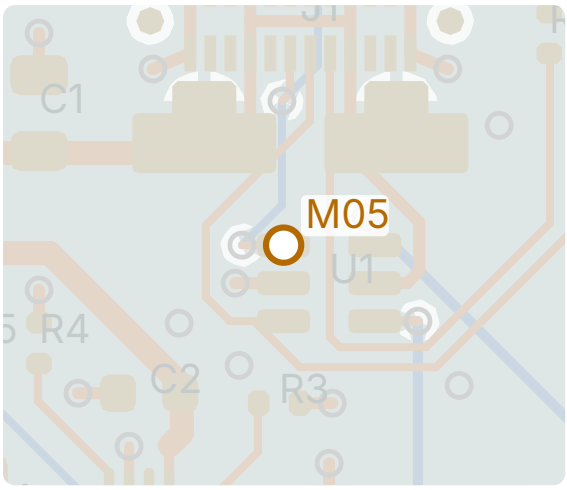
M02: exposed_pad_vias —



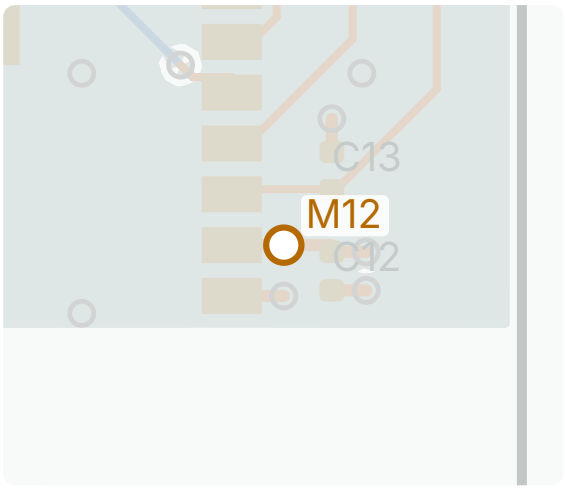
M03: edge_clearance —



M04: acute_angles —

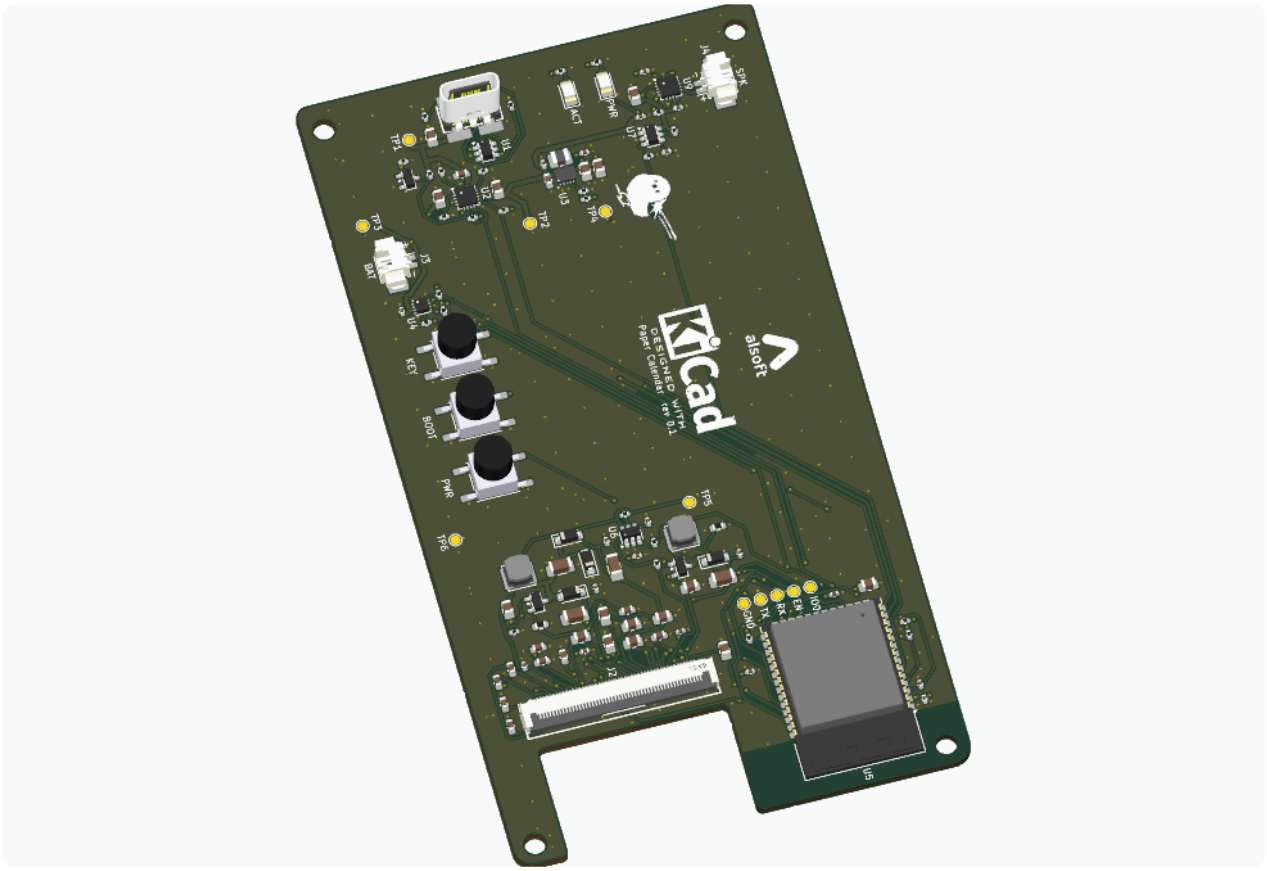


M05: esd_order —



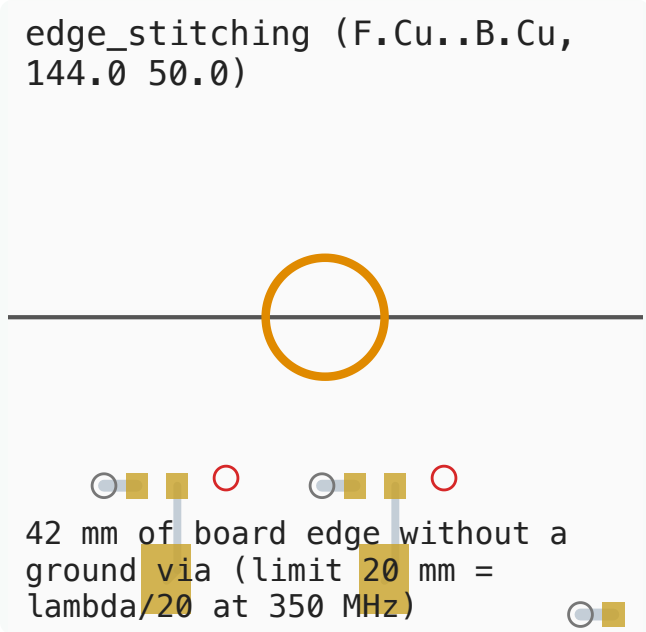
M12: current_width —

6. 3D render

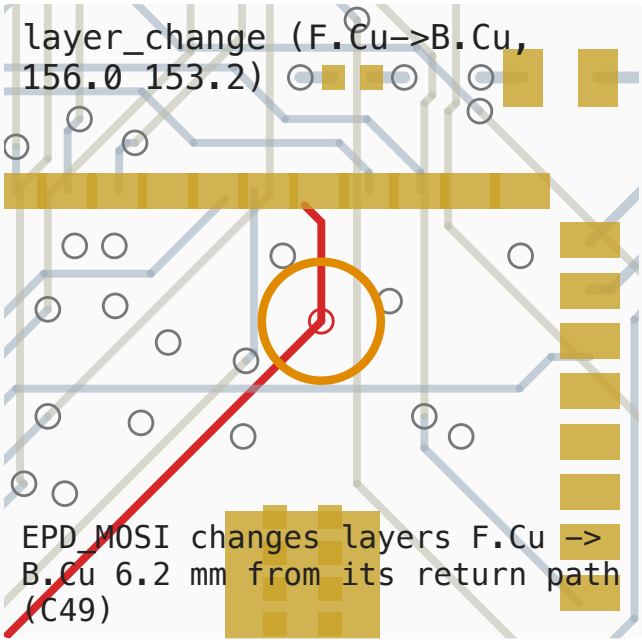


3D board render (available models only)

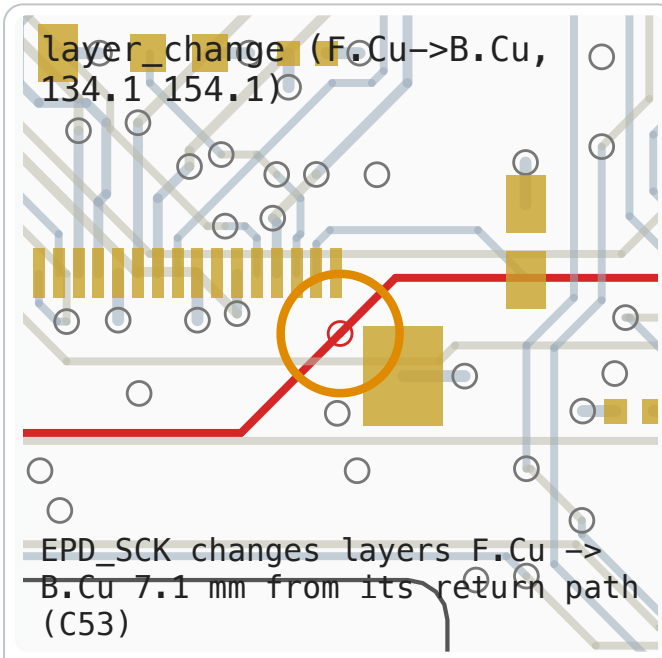
7. EMC maps



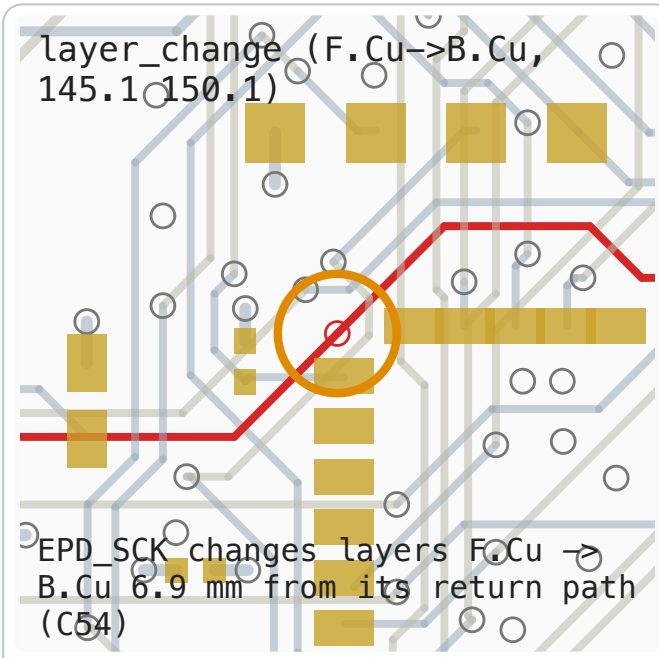
emc: artifacts/emc-0001.svg



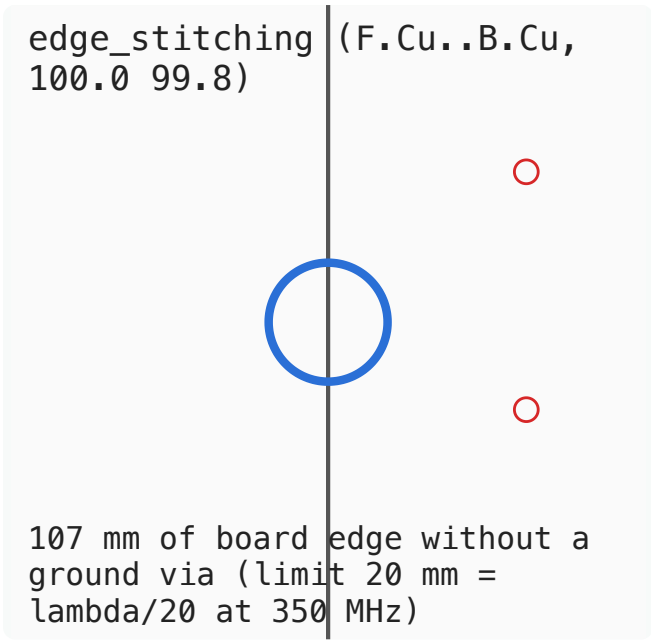
emc: artifacts/emc-0002.svg



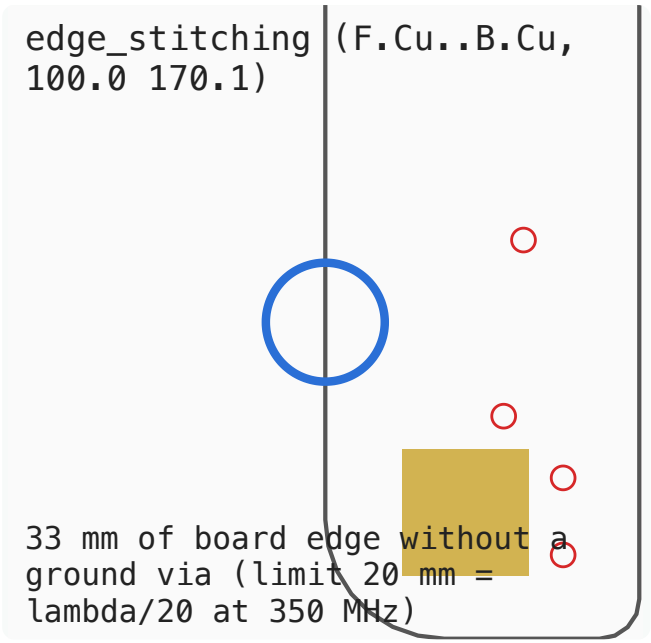
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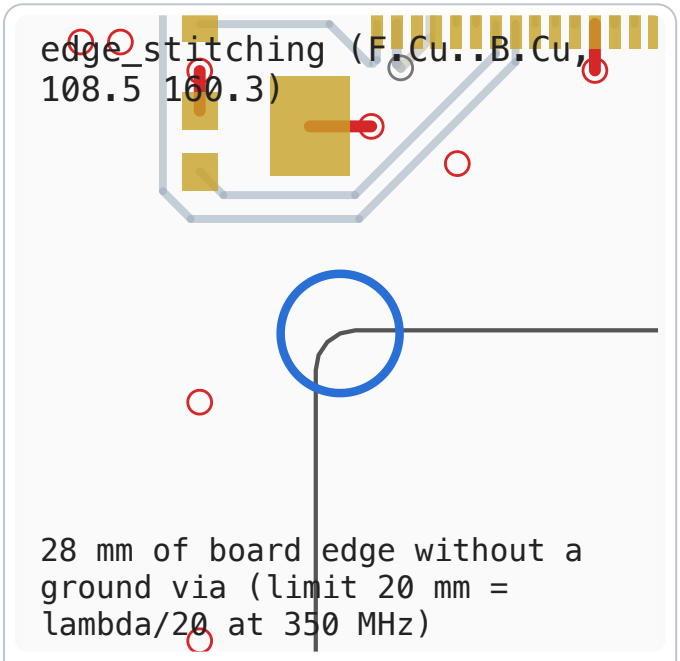
emc: artifacts/emc-0004.svg



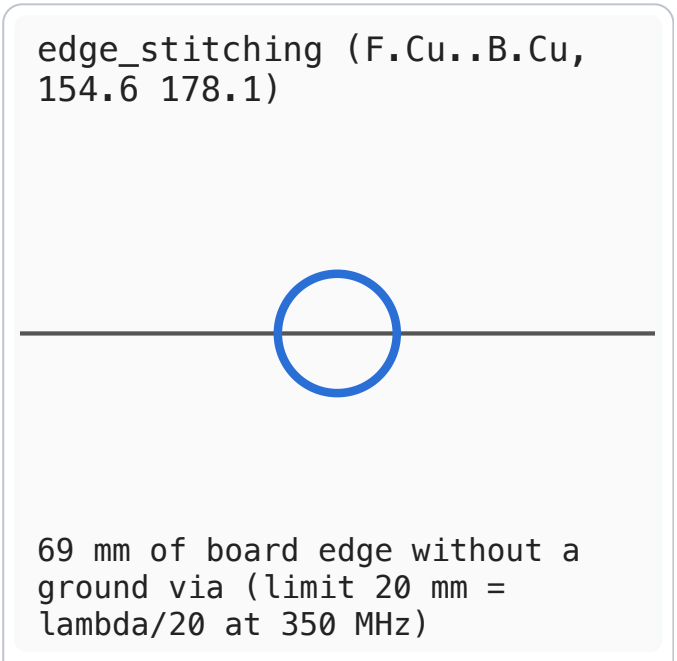
emc: artifacts/emc-0005.svg



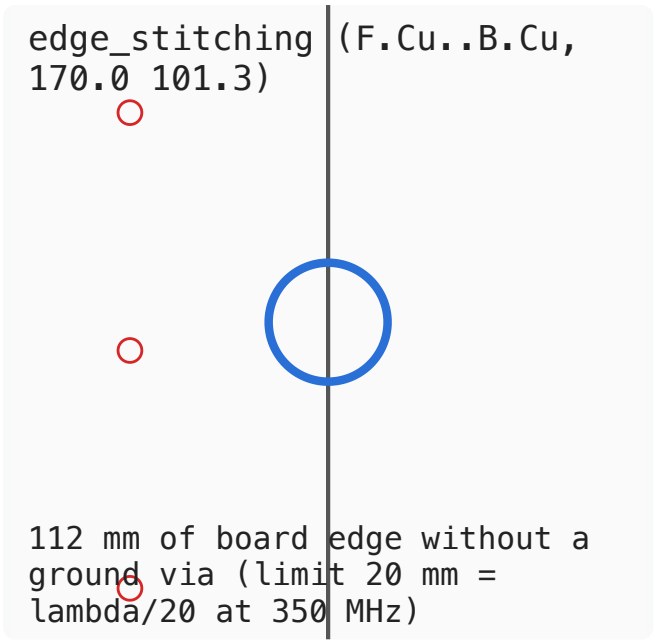
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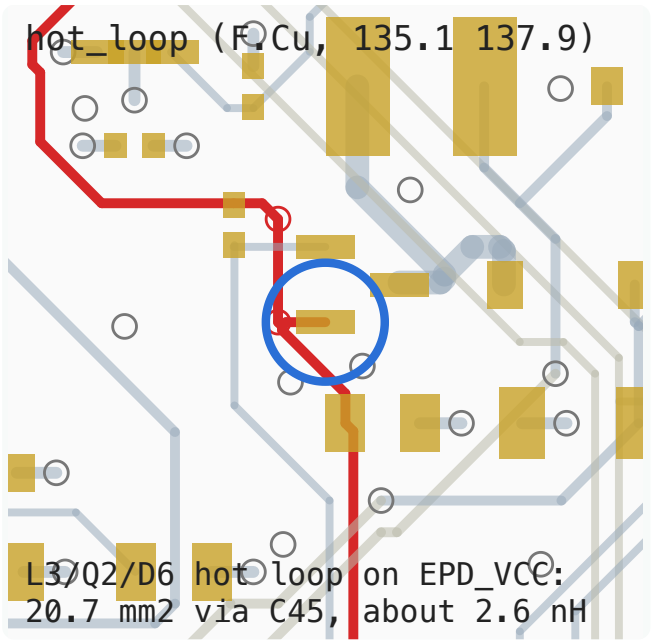
emc: artifacts/emc-0007.svg



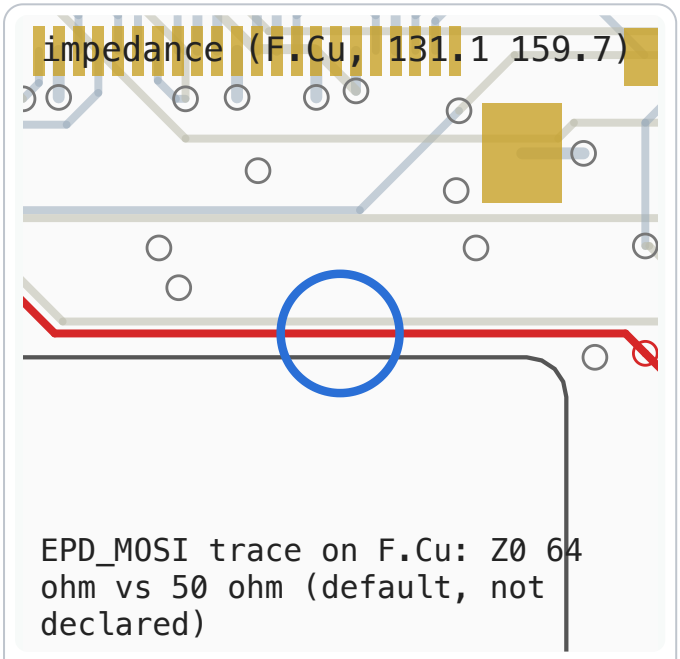
emc: artifacts/emc-0008.svg



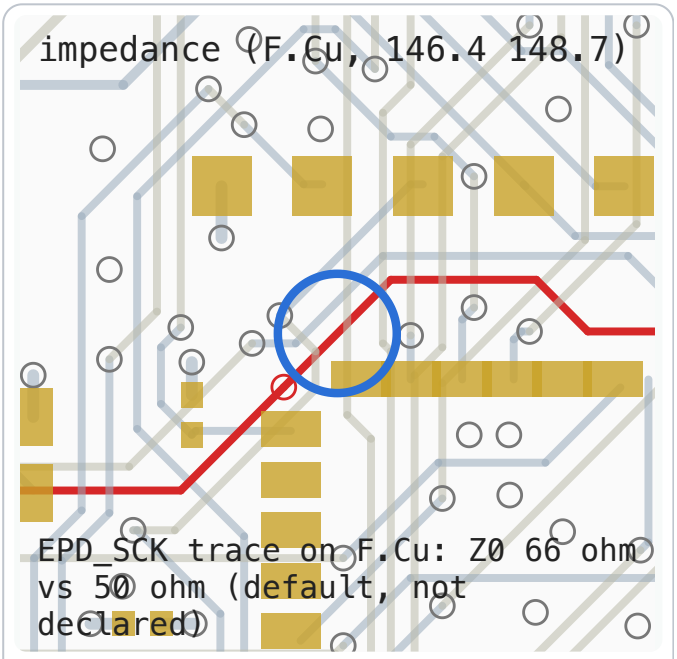
emc: artifacts/emc-0009.svg



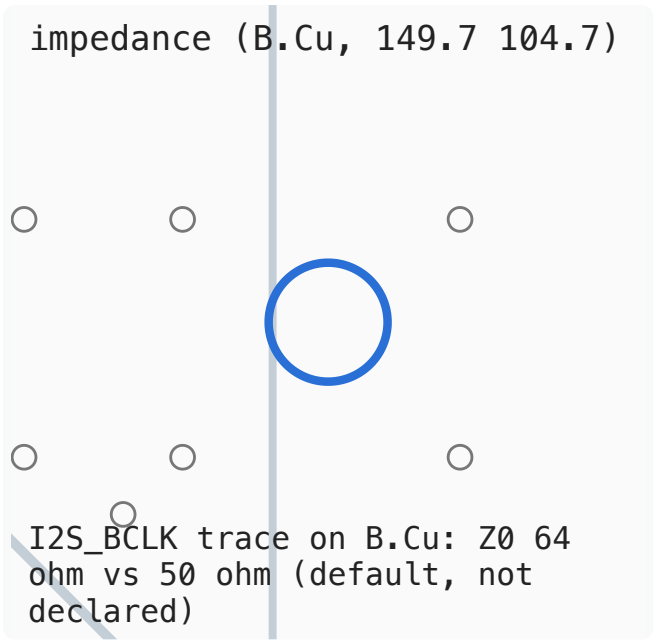
emc: artifacts/emc-0010.svg



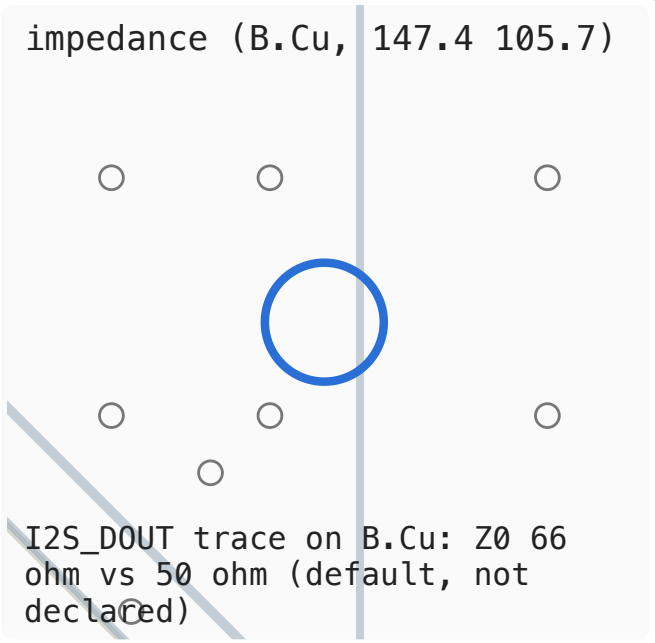
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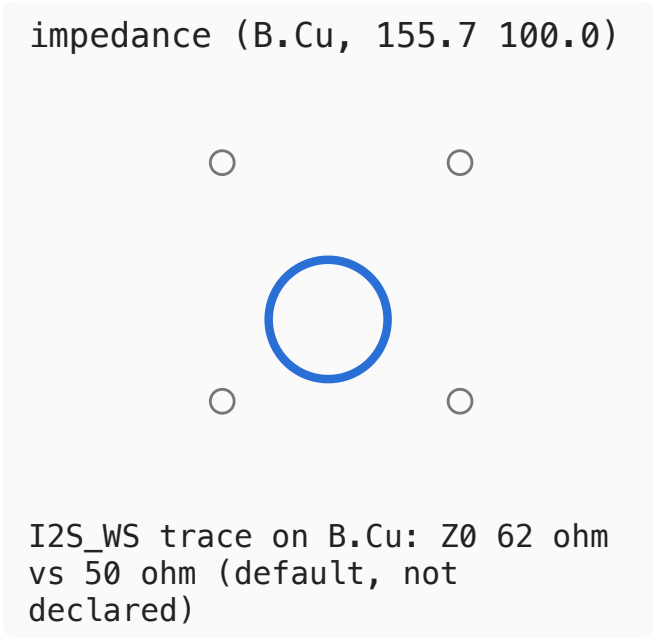
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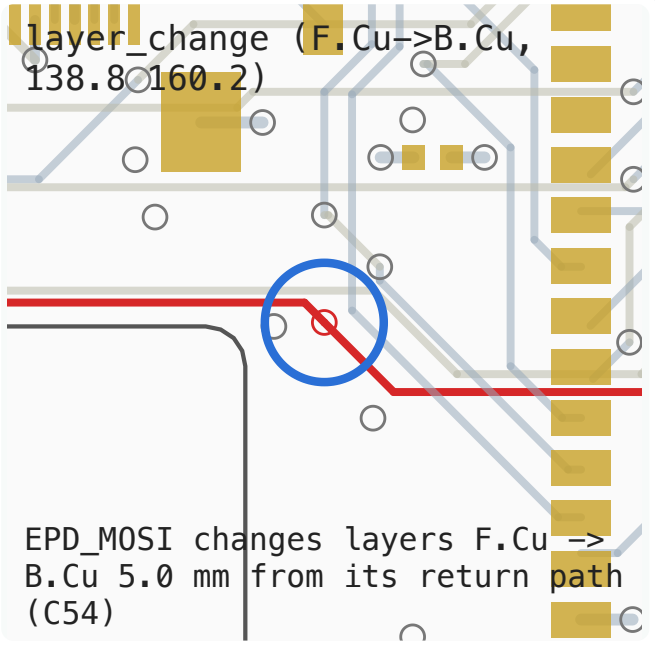
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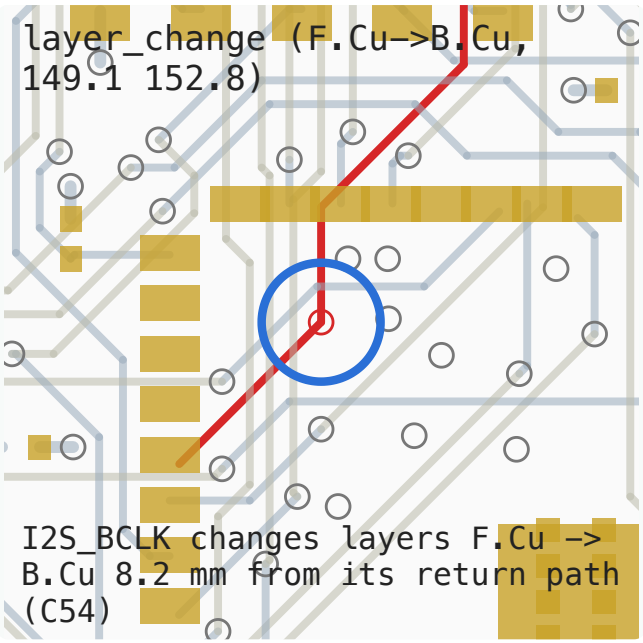
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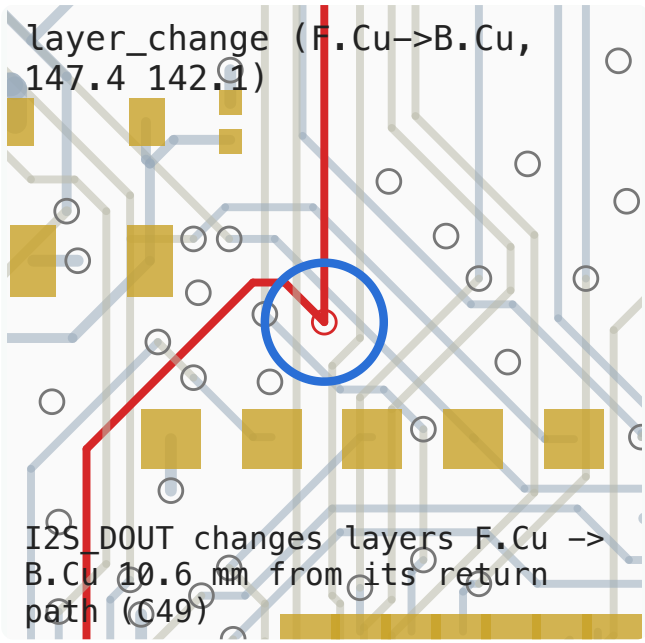
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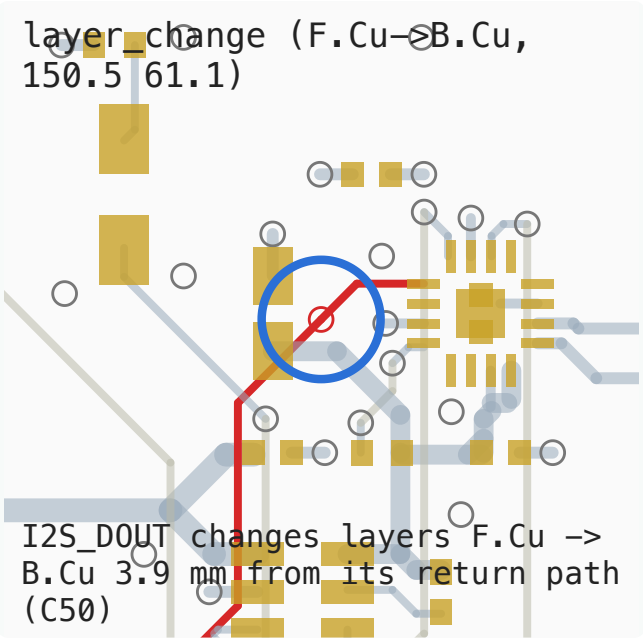
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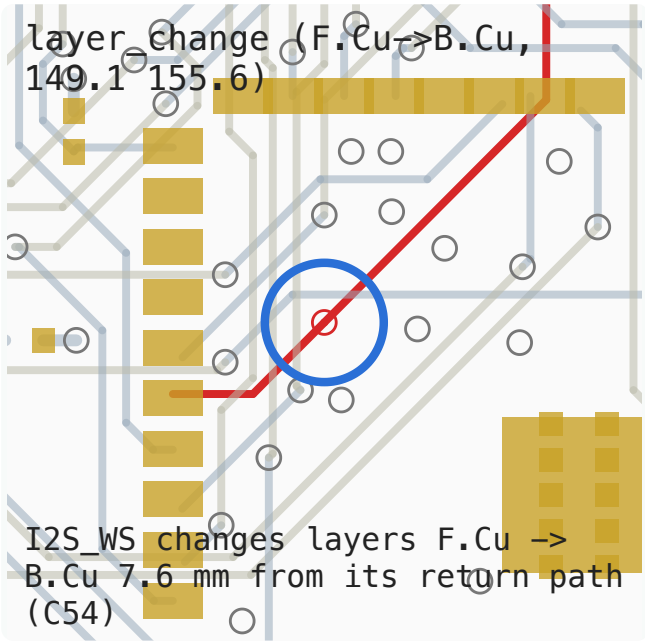
emc: artifacts/emc-0017.svg



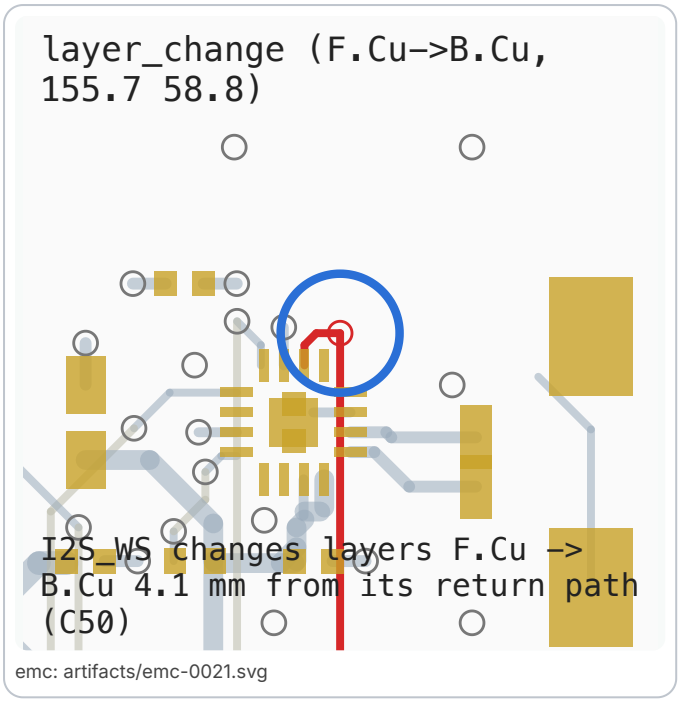
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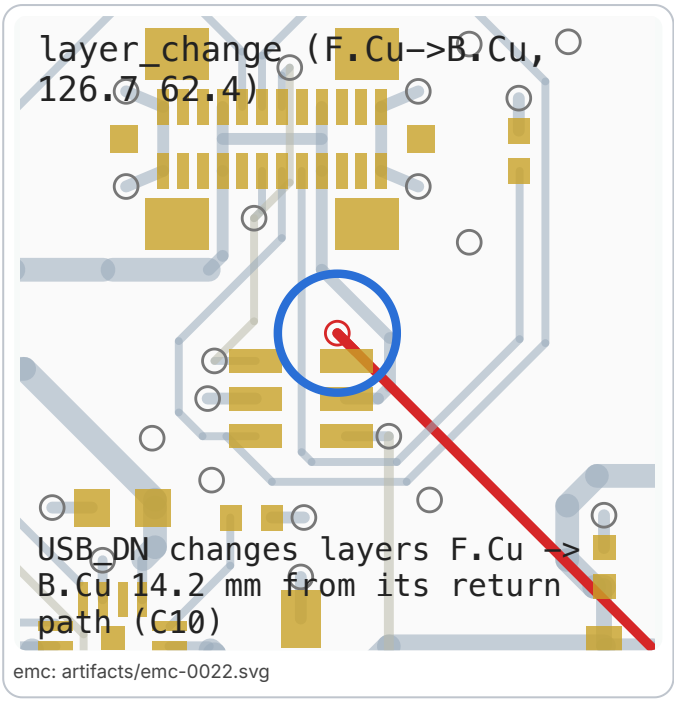
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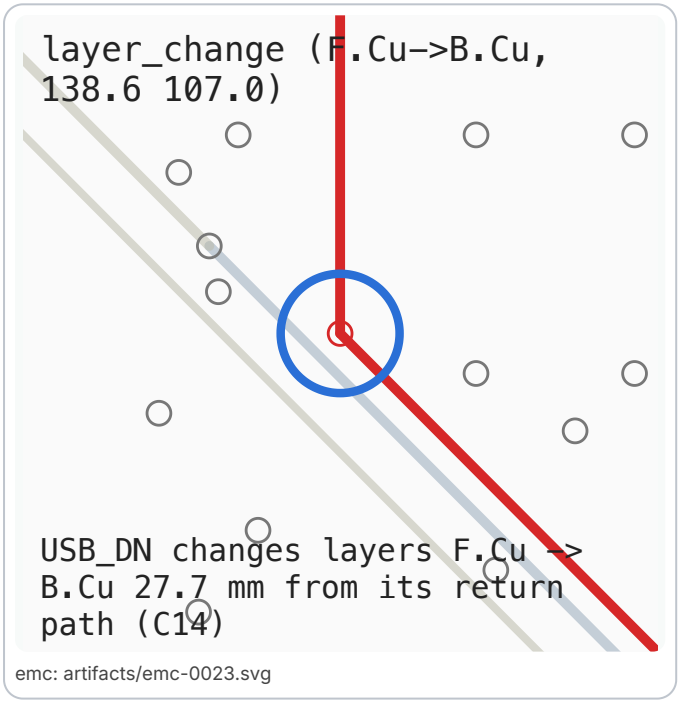
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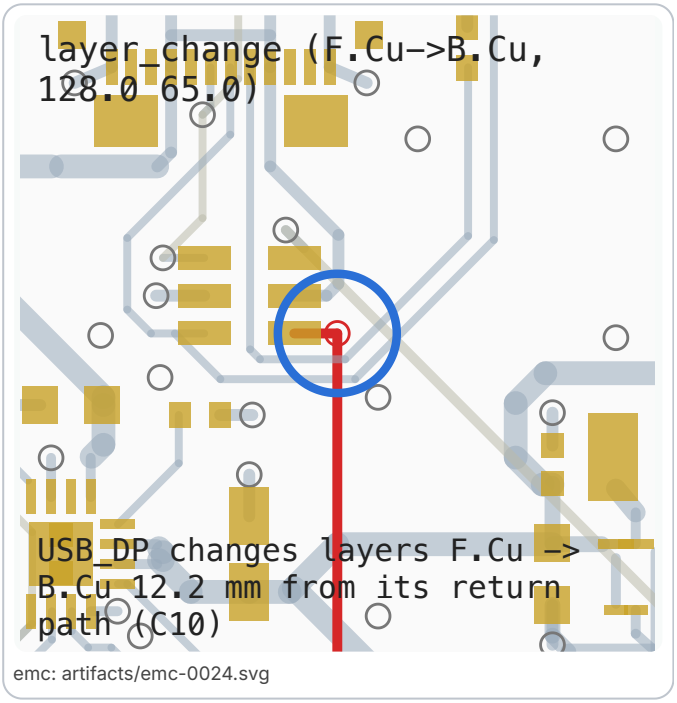
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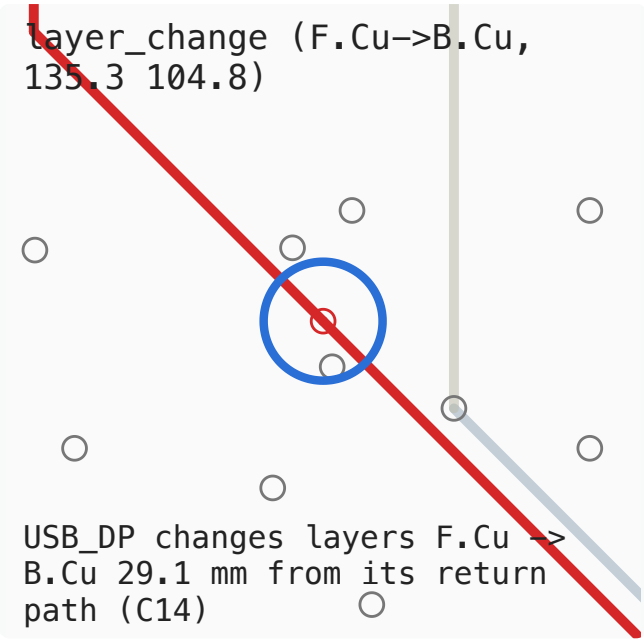
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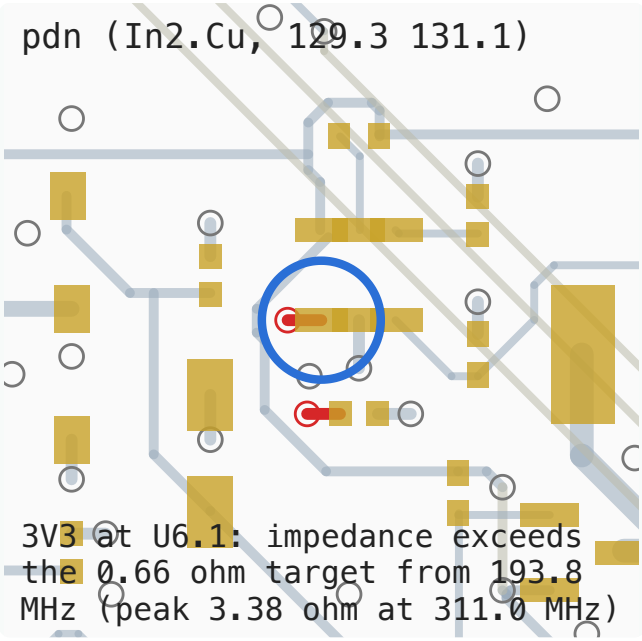
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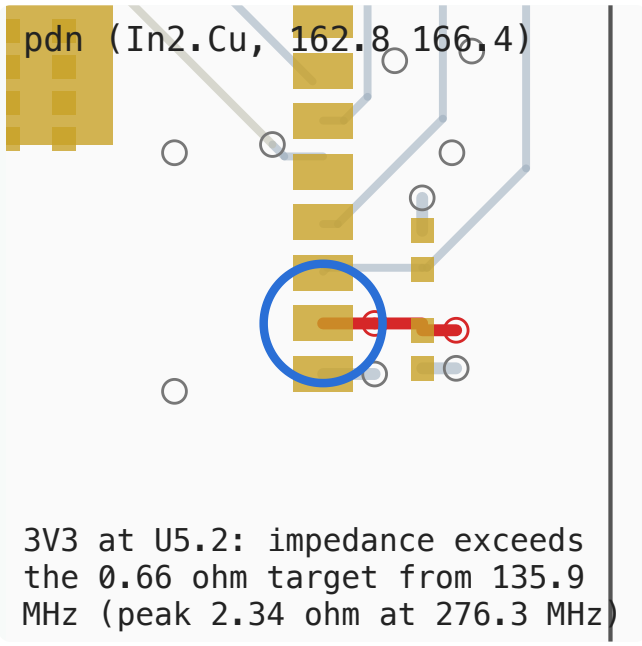
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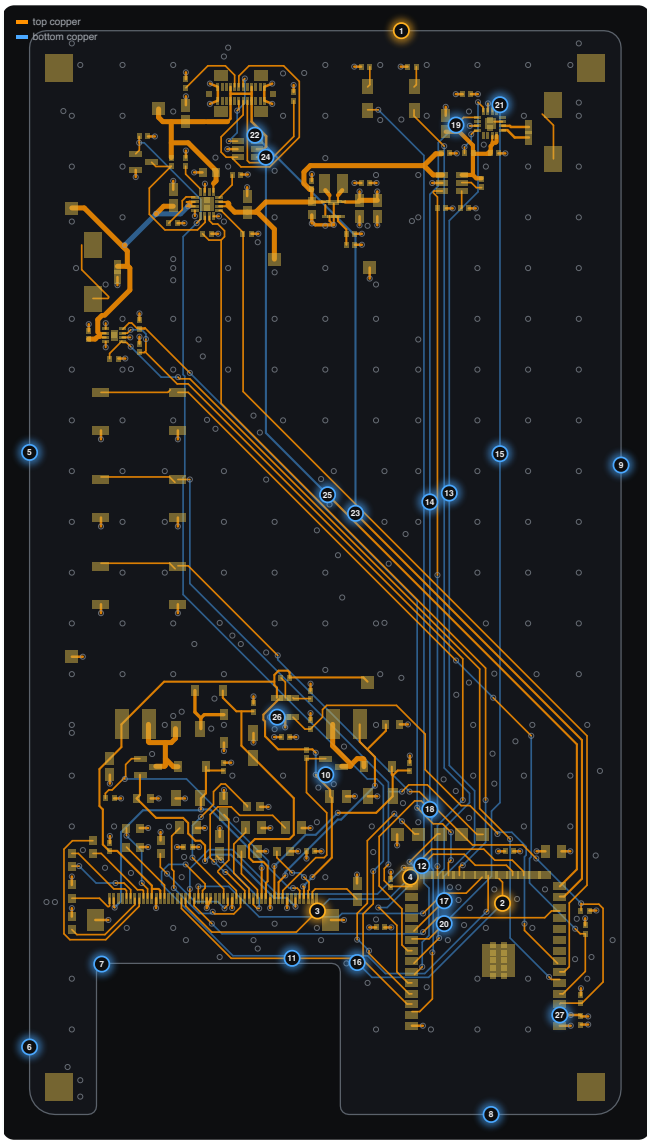
emc: artifacts/emc-0025.svg



emc: artifacts/emc-0026.svg

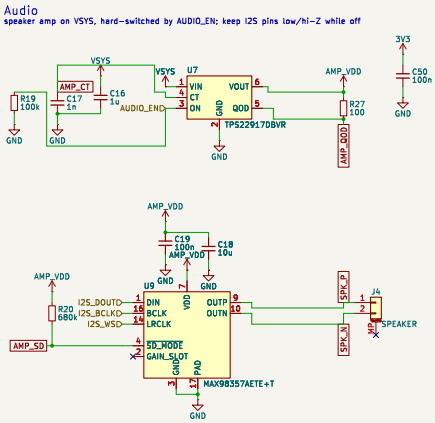


emc: artifacts/emc-0027.svg

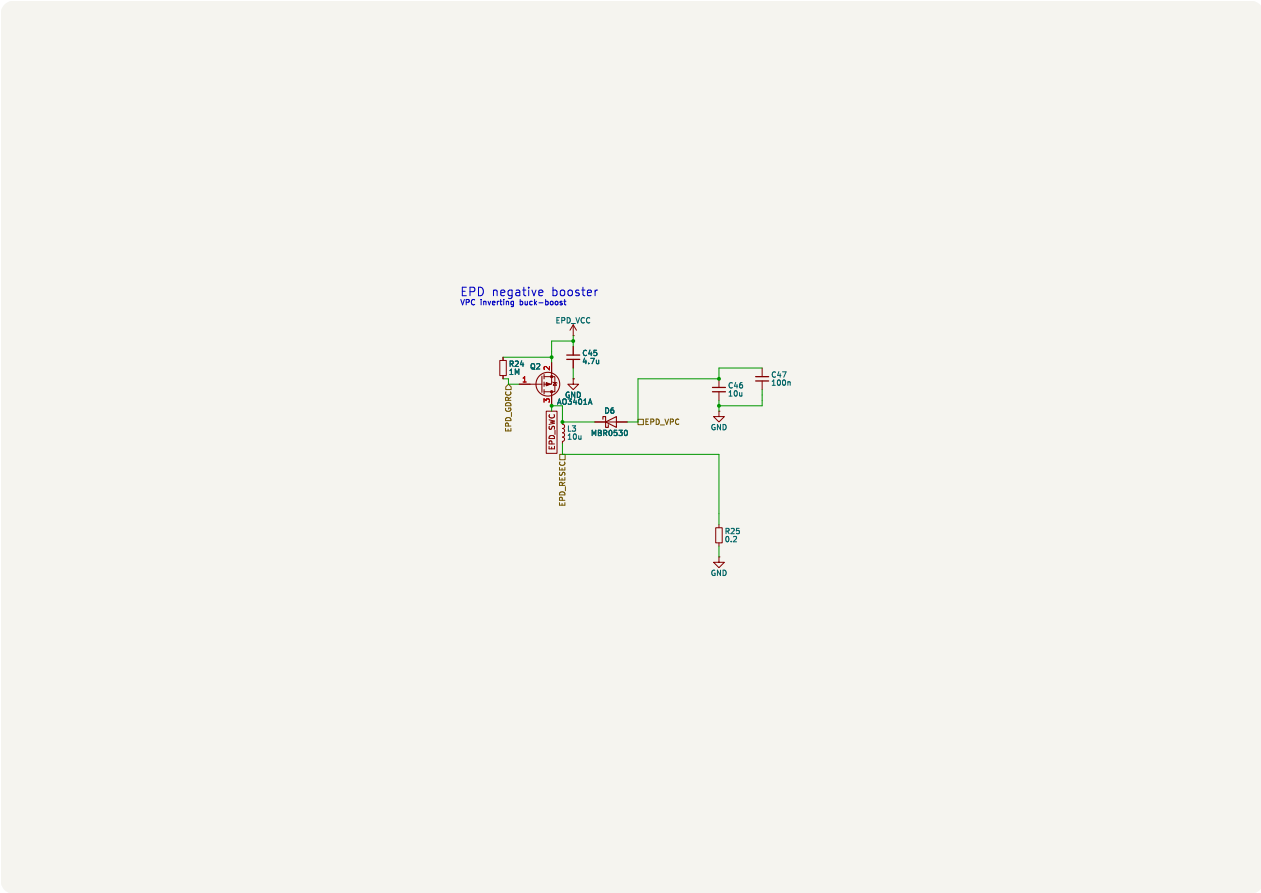


emc: artifacts/emc-map.svg

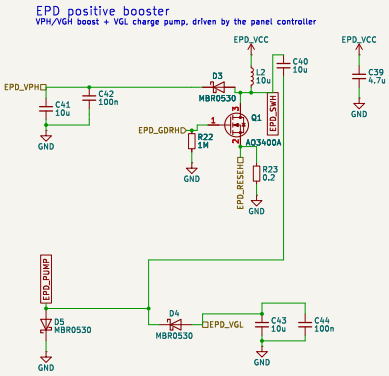
8. Schematic excerpts



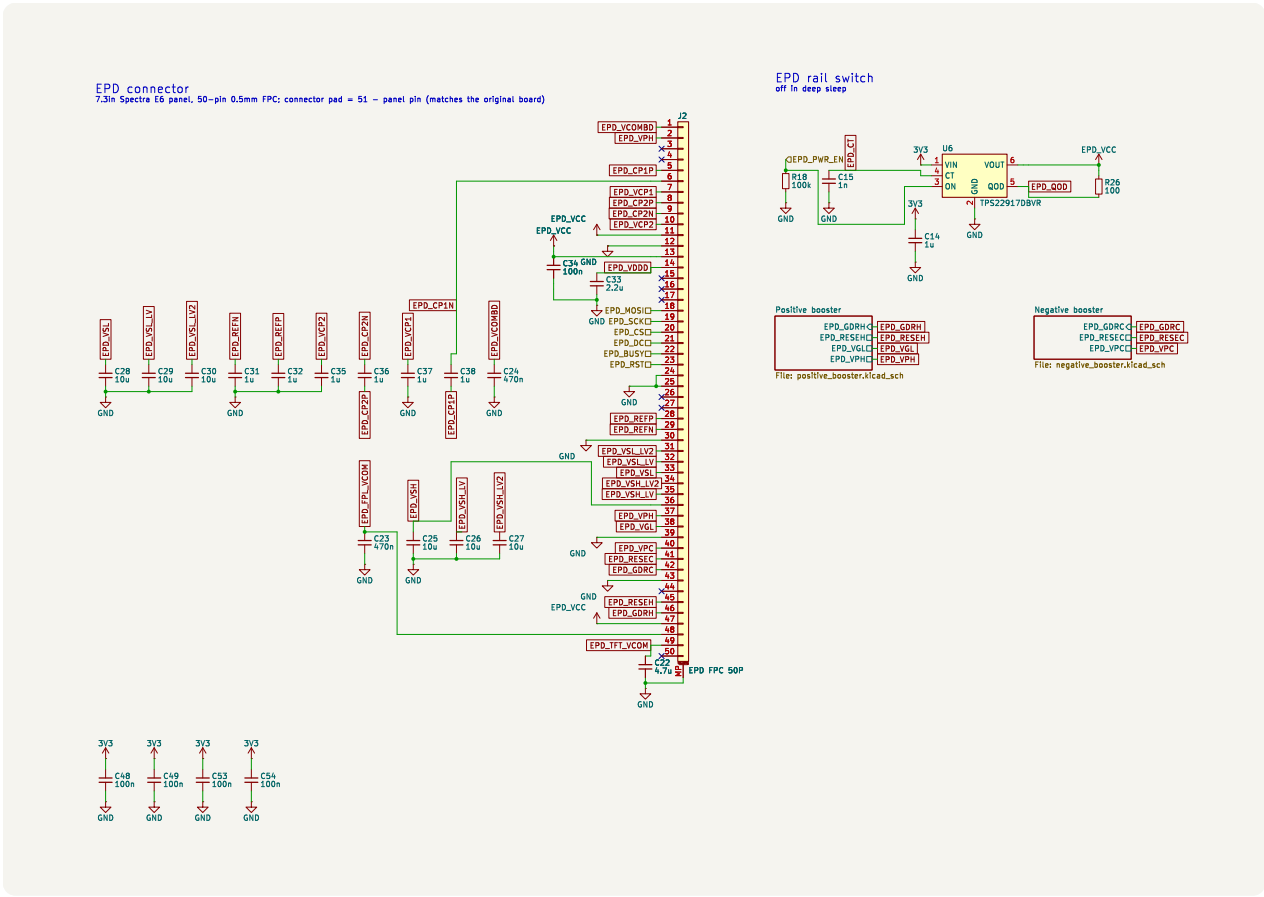
Schematic excerpt: paper_calendar-Audio.svg



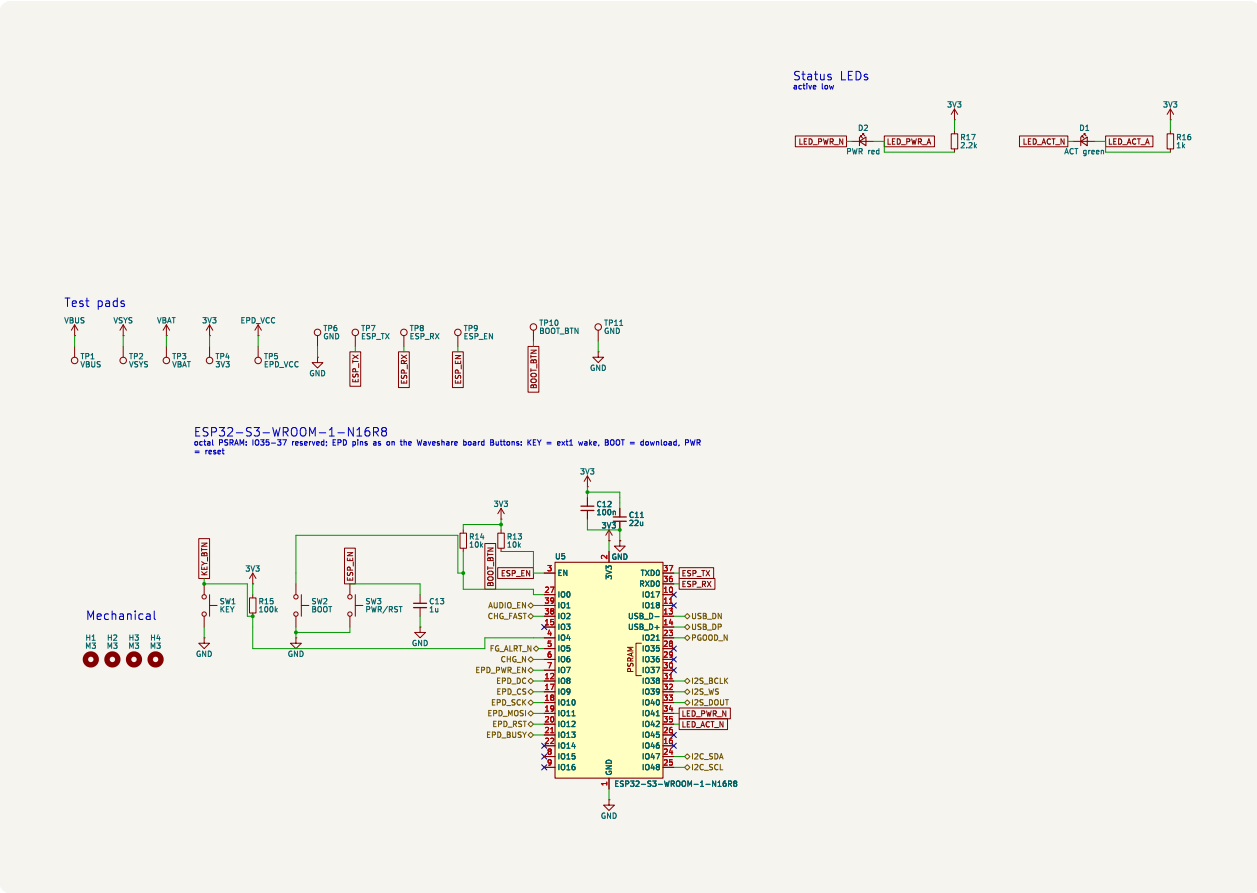
Schematic excerpt: paper_calendar-EPD-Negative booster.svg



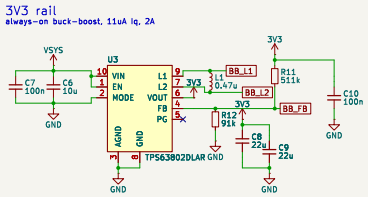
Schematic excerpt: paper_calendar-EPD-Positive booster.svg



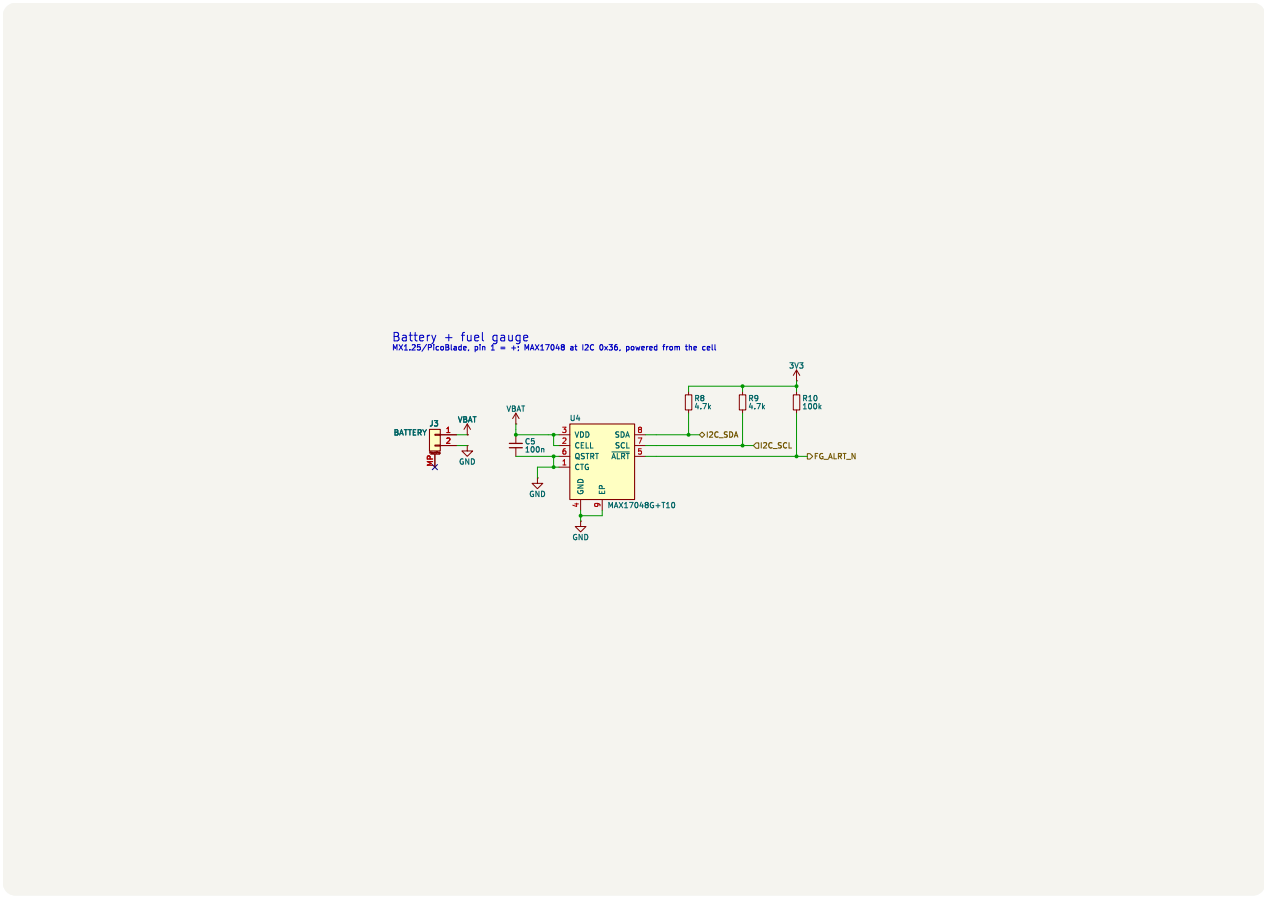
Schematic excerpt: paper_calendar-EPD.svg



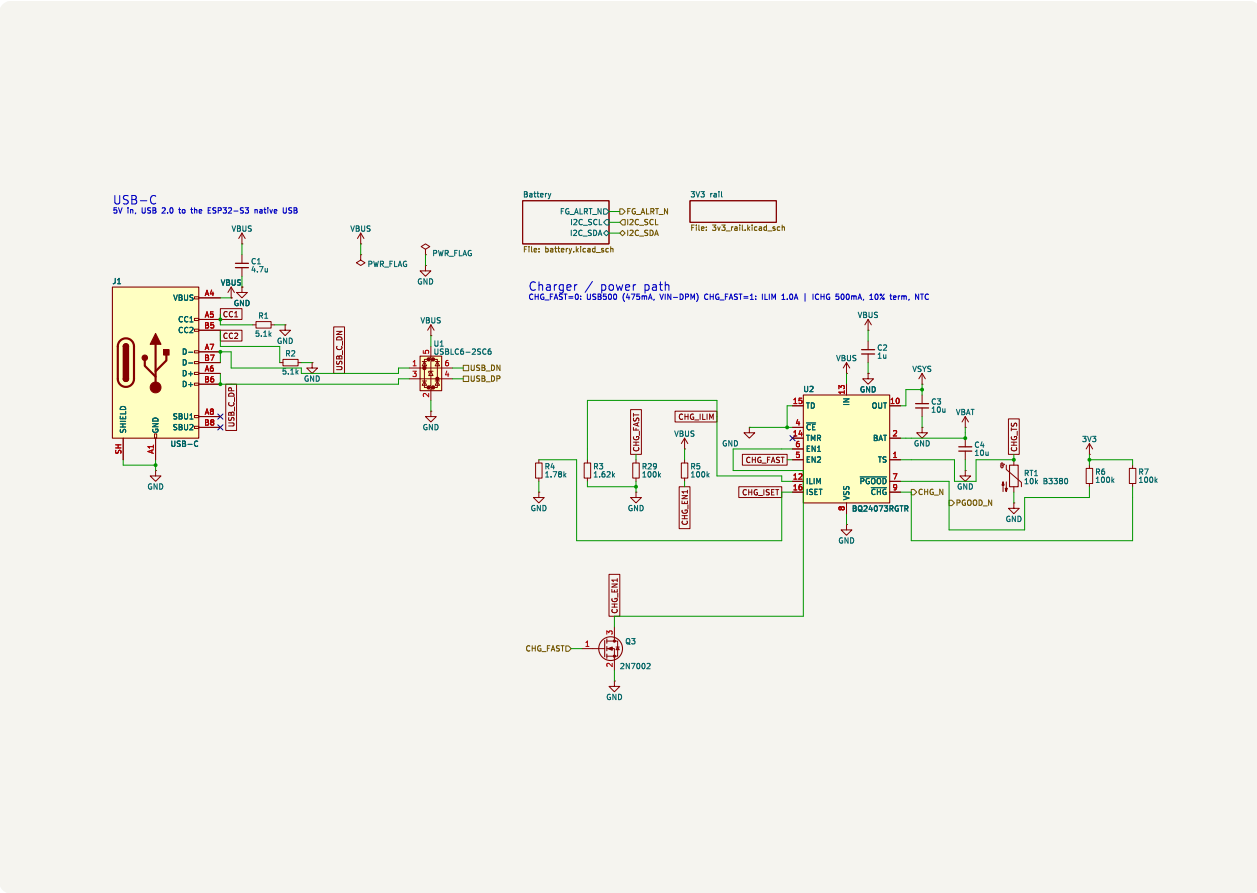
Schematic excerpt: paper_calendar-MCU.svg



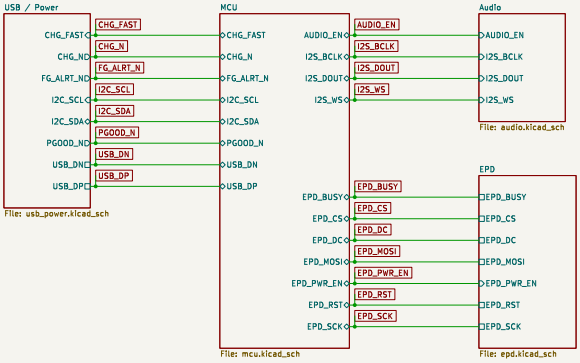
Schematic excerpt: paper_calendar-USB _ Power-3V3 rail.svg



Schematic excerpt: paper_calendar-USB _ Power-Battery.svg



Schematic excerpt: paper_calendar-USB _ Power.svg



Schematic excerpt: paper_calendar.svg

9. Firmware / hardware table

Step	Result	Evidence / limitation	Findings
firmware	incomplete	bringup/bringup.py: unsupported source language; provide a hardware-layer adapter; bringup/link.py: unsupported source language; provide a hardware-layer adapter; bringup/ui.py: unsupported source language; provide a hardware-layer adapter; scripts/dither.py: unsupported source l... (full evidence in appendix)	

No additional visual evidence supplied by these steps.

10. Emulation results

Step	Result	Evidence / limitation	Findings
emulation	incomplete	review-miss checks: startup: "J1" needs r_series with a cited basis; startup: "J3" needs r_series with a cited basis; startup: U5 3V3: load lacks minimum operating-voltage evidence; startup: J2 EPD_VCC: load lacks minimum operating-voltage evidence; startup: U9 AMP_VDD: load lack... (full evidence in appendix)	I35, I36, I37, I38, I39, I40, I41, I42

I35: MCU scenario first_boot: passed — see the finding card and Appendix A.

I36: MCU scenario second_wake: passed — see the finding card and Appendix A.

I37: MCU scenario key_wake: passed — see the finding card and Appendix A.

I38: MCU scenario key_pressed_during_wake: passed — see the finding card and Appendix A.

I39: MCU scenario low_battery: passed — see the finding card and Appendix A.

I40: MCU scenario router_off: passed — see the finding card and Appendix A.

I41: MCU scenario night_no_wifi: passed — see the finding card and Appendix A.

I42: Digital emulation scope — see the finding card and Appendix A.

No additional visual evidence supplied by these steps.

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11. Bring-up simulation results

Step	Result	Evidence / limitation	Findings
bringup	incomplete	Read 29 supplied check/prompt expressions and their constants, source lines and SHA-256 hashes in hardware/build/bringup-source.json. Awaiting real serial responses and operator answers to the retained source prompts. Host predicates have not been executed; replay against an MCU ... (full evidence in appendix)	I09

I09: Bring-up source inventory — see the finding card and Appendix A.

No additional visual evidence supplied by these steps.

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12. Datasheet limits and checks

Step	Result	Evidence / limitation	Findings
intent	incomplete	3V3: current unresolved; schematic net intent: ; net name 3V3; nominal only, tolerance unspecified; unresolved demand for R10, R11, R13, R14, R15, R16, R17, R6, R7, R8, R9, U6; known subtotal 0.7 A: U5 declared load 0.5 A; U6 input: declared downstream source/load subtotal 0.2 A;… (full evidence in appendix)	
datasheets	incomplete	J1: download is not PDF data (missing %PDF- header); supply a direct PDF URL; U1: https://www.st.com/resource/en/datasheet/usblc6-2.pdf: datasheet tool failed (exit 28): curl: (28) Operation timed out after 40011 milliseconds with 0 bytes received (host www.st.com stalls or reset… (full evidence in appendix)	
ratings	incomplete	C1 effective_fraction: need exactly one applicable extracted row, found 0; C1 land_pad_height: pad 1 land pattern not independently extracted; C1 land_pad_height: pad 2 land pattern not independently extracted; C1 land_pad_width: pad 1 land pattern not independently extracted; C1… (full evidence in appendix)	I48, I49, I50, I51, I52, I53

I48: powerpath coverage — see the finding card and Appendix A.

I49: open_drain_pullup — see the finding card and Appendix A.

I50: open_drain_pullup — see the finding card and Appendix A.

I51: open_drain_pullup — see the finding card and Appendix A.

I52: open_drain_pullup — see the finding card and Appendix A.

I53: open_drain_pullup — see the finding card and Appendix A.

No additional visual evidence supplied by these steps.

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13. Review steps

Step	Ran	Why / coverage
import	true	Completed its available checks; assumptions remain in source artifacts.
intent	false	3V3: current unresolved; schematic net intent: ; net name 3V3; nominal only, tolerance unspecified; unresolved demand for R10, R11, R13, R14, R15, R16, R17, R6, R7, R8, R9, U6; known subtotal 0.7 A: U5 declared load 0.5 A; U6 input: declared downstream source/load subtotal 0.2 A;... (full evidence in appendix)
datasheets	false	J1: download is not PDF data (missing %PDF- header); supply a direct PDF URL; U1: https://www.st.com/resource/en/datasheet/usblc6-2.pdf: datasheet tool failed (exit 28): curl: (28) Operation timed out after 40011 milliseconds with 0 bytes received (host www.st.com stalls or reset... (full evidence in appendix)
ratings	false	C1 effective_fraction: need exactly one applicable extracted row, found 0; C1 land_pad_height: pad 1 land pattern not independently extracted; C1 land_pad_height: pad 2 land pattern not independently extracted; C1 land_pad_width: pad 1 land pattern not independently extracted; C1... (full evidence in appendix)
board	true	Completed its available checks; assumptions remain in source artifacts.
emc	true	Completed its available checks; assumptions remain in source artifacts.
current	false	Incomplete copper intent coverage: current_uncovered_nets: 77 routed nets lacking current demand; voltage_uncovered_nets: 77 routed nets lacking peak voltage intervals

erules	false	review-miss checks: open_drain_source: missing datasheet pin_type for U1.1, U1.3, U1.4, U1.6, U2.1, U2.12, U2.16, Q3.3, U3.9, U3.7, U9.17, Q1.2, Q1.3, D3.1, D3.2, D4.1, D4.2, D5.2, Q2.3, D6.1, D6.2
drc	true	Completed its available checks; assumptions remain in source artifacts.
erc	true	Completed its available checks; assumptions remain in source artifacts.
firmware	false	bringup/bringup.py: unsupported source language; provide a hardware-layer adapter; bringup/link.py: unsupported source language; provide a hardware-layer adapter; bringup/ui.py: unsupported source language; provide a hardware-layer adapter; scripts/dither.py: unsupported source l... (full evidence in appendix)
emulation	false	review-miss checks: startup: "J1" needs r_series with a cited basis; startup: "J3" needs r_series with a cited basis; startup: U5 3V3: load lacks minimum operating-voltage evidence; startup: J2 EPD_VCC: load lacks minimum operating-voltage evidence; startup: U9 AMP_VDD: load lack... (full evidence in appendix)
bringup	false	Read 29 supplied check/prompt expressions and their constants, source lines and SHA-256 hashes in hardware/build/bringup-source.json. Awaiting real serial responses and operator answers to the retained source prompts. Host predicates have not been executed; replay against an MCU ... (full evidence in appendix)
requirements	true	Completed its available checks; assumptions remain in source artifacts.
simulation	true	Completed its available checks; assumptions remain in source artifacts.
report	true	Completed its available checks; assumptions remain in source artifacts.

14. Not checkable without hardware

Physical EMC emissions/immunity, thermal rise, real component tolerances, assembly workmanship and connector/mechanical fit require prototype measurements. These are hardware obligations; missing software steps are listed under Review steps and Tool limitations.

Obligation	State
EMC emissions and immunity	needs-hardware
Thermal rise and component tolerances	needs-hardware
Assembly workmanship and mechanical fit	needs-hardware

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15. Tool limitations

CAD checks and estimates do not establish certification. A 3D render uses available models only. Schematic excerpts generated from converted connectivity do not preserve the original drawing. Derived values require reviewer approval at their cited sources.

intent: 3V3: current unresolved; schematic net intent: ; net name 3V3; nominal only, tolerance unspecified; unresolved demand for R10, R11, R13, R14, R15, R16, R17, R6, R7, R8, R9, U6; known subtotal 0.7 A: U5 declared load 0.5 A; U6 input: declared downstream source/load subtotal 0.2 A;… (full evidence in appendix)

datasheets: J1: download is not PDF data (missing %PDF- header); supply a direct PDF URL; U1: https://www.st.com/resource/en/datasheet/usblc6-2.pdf: datasheet tool failed (exit 28): curl: (28) Operation timed out after 40011 milliseconds with 0 bytes received (host www.st.com stalls or reset… (full evidence in appendix)

ratings: C1 effective_fraction: need exactly one applicable extracted row, found 0; C1 land_pad_height: pad 1 land pattern not independently extracted; C1 land_pad_height: pad 2 land pattern not independently extracted; C1 land_pad_width: pad 1 land pattern not independently extracted; C1… (full evidence in appendix)

current: Incomplete copper intent coverage: current_uncovered_nets: 77 routed nets lacking current demand; voltage_uncovered_nets: 77 routed nets lacking peak voltage intervals

erules: review-miss checks: open_drain_source: missing datasheet pin_type for U1.1, U1.3, U1.4, U1.6, U2.1, U2.12, U2.16, Q3.3, U3.9, U3.7, U9.17, Q1.2, Q1.3, D3.1, D3.2, D4.1, D4.2, D5.2, Q2.3, D6.1, D6.2

firmware: bringup/bringup.py: unsupported source language; provide a hardware-layer adapter; bringup/link.py: unsupported source language; provide a hardware-layer adapter; bringup/ui.py: